$^{34}S(d,n)$ **1969Da12**

$J^{\pi}=0^{+}$ for ^{34}S ground state.

1969Da12: a 4.95-MeV deuteron beam was produced from the 5.5-MeV Van de Graaff accelerator at University of Alberta and impinged on a CdS target enriched to 37% ^{34}S evaporated onto a 0.05-cm gold backing. Neutrons were detected by NE213 scintillators using the time-of-flight (tof) method with FWHM=1.0 ns. Measured $\sigma(E_n,\theta)$. Deduced levels, J, π, and L-transfers from DWUCK-DWBA analysis of the measured $\sigma(\theta)$.

1993TeZX: $^{34}S(d,n)^{35}Cl$ at E(d)=25 MeV at CYRJC, Tohoku University.

^{35}Cl Levels

E(level)†	L‡	Comments
0	2	
1220	0	
1762	2	
2645	(1)	L: weak stripping or possibly two unresolved levels.
2695		
3006	3	
3163	3	L: Discrepancy: L=2 from 0^{+} in (^{3}He,d), (d,^{3}He), and (pol d,^{3}He).
4060		
4110		
4170		

† From 1969Da12.
‡ From DWBA analysis of the measured $\sigma(\theta)$ in 1969Da12.